

Savitzky-Golay optimizer diagnostics

Objective: mean squared translational/angular acceleration residual; angular residual uses the reference inertia/mass metric

Tolerances: {'ftol': None, 'xtol': None, 'gtol': 1e-06, 'note': 'ftol and xtol are disabled by default in the SG estimator; gtol or max_nfev terminates the solve unless explicitly overridden'}

Selected strict-ZOH solve

status=1 success=True nfev=4 njev=4

message='gtol' termination condition is satisfied.

reported optimality=2.3125099168908037e-08

initial={'cost': 1.4248742718942367, 'gradient_inf_norm': 0.021572309760144126, 'gradient_l2_norm': 0.024830486085077413, 'jacobian': {'condition_number_nonzero_subspace': 409.9536688423053, 'numerical_rank': 14, 'numerical_tolerance': 1.0479648320947052e-10, 'singular_values': [119.12200977584803, 117.86756634077359, 102.90165180585939, 48.5991493897313, 44.726391565282306, 25.868650569998067, 12.989364429433579, 11.738638047154252, 11.31168879330844, 11.042247827010474, 9.896579427464276, 3.7985560998250185, 3.7925582437115826, 0.29057432297714125]}, 'residual_l2_norm': 1.6881198250682543}

final={'cost': 1.4248742280551663, 'gradient_inf_norm': 2.3125099168908037e-08, 'gradient_l2_norm': 2.8017349687240318e-08, 'jacobian': {'condition_number_nonzero_subspace': 409.998268653704, 'numerical_rank': 14, 'numerical_tolerance': 1.0479648613947692e-10, 'singular_values': [119.12201310638204, 117.8676331296809, 102.901663331624, 48.59915723838814, 44.72639054474949, 25.868651609928396, 12.989357053587765, 11.738639637341667, 11.311686020520677, 11.042253998544393, 9.896577337204729, 3.7985592788612026, 3.792555129543104, 0.2905427222840198]}, 'residual_l2_norm': 1.6881197990990842}

coordinate_step_l2_norm=5.7280483137447645e-05

cost_reduction=4.3839070373863365e-08

gradient_tolerance_satisfied=True

finite_difference_max_relative_error=9.175095543339716e-12

finite_difference_directional_checks=[{'analytic_l2_norm': 108.98780029913112, 'difference_l2_norm': 9.999734808040419e-10, 'direction': 'negative_gradient', 'finite_difference_l2_norm': 108.98780029924937, 'relative_error': 9.175095543339716e-12, 'scheme': 'central', 'step': 6.0554544523933395e-06}]

All smoothstep and strict-ZOH stage details are in optimizer_diagnostics.json.